

Wenbo Lyu

B.Sc. Graduate

Curriculum Vitae

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Education and qualifications

B.Sc. In Geographic Information Science Shaanxi Normal University 2025

Research Experience

2026.6- Research Assistant, The Hong Kong Polytechnic University
2025.9-2026.6 Research Assistant, Eastern Institute of Technology, Ningbo
2024.8-2025.8 Research Intern, The Hong Kong University of Science and Technology (Guangzhou)

Research interests

- My research interests lie in **advancing methodologies in spatial causal inference** and **developing high-performance computational tools**, with a primary focus on *R packages*.
- Currently, my work centers on **Empirical Dynamic Modeling (EDM)** framework for modeling *dynamic system*, **information theory** for quantifying *information flow* and *variable interactions*, **ordinary differential equations (ODEs)** for characterizing *spatiotemporal processes* and *system evolution*, as well as **counterfactual** and **potential outcomes** framework for estimating *causal effects*. I am particularly interested in leveraging these approaches to address critical challenges in *urban sustainability*, *climate change mitigation*, and broader global issues.
- I specialize in *data analysis*, *statistical modeling*, and developing open-source analytical tools, including *R packages*, using **R**, **C++**, and **Python**, with a strong focus on *spatial analysis*. I actively contribute to the R geospatial community and am dedicated to advancing open-source geospatial software.

Selected research papers

- ¹W. Lyu, S. Dai, Y. Song, W. Zhao, W. Yi, Y. Xiao, and N. Jia, "Measuring causal strengths from spatial cross-sectional data with geographical cross mapping cardinality", *International Journal of Geographical Information Science*, 1–23 (2026).
- ²W. Lyu, Y. Lei, W. Yi, Y. Song, X. Li, S. Dai, Y. Qin, and W. Zhao, "Causal discovery in urban data with temporal empirical dynamic modeling: the R package tEDM", *Computers, Environment and Urban Systems* **127**, 102435 (2026).
- ³W. Lv, Y. Lei, F. Liu, J. Yan, Y. Song, and W. Zhao, "gdverse: an R package for spatial stratified heterogeneity family", *Transactions in GIS* **29**, e70032 (2025).
- ⁴W. Lv, F. Liu, K. Cai, Y. Cao, M. Deng, W. Liang, J. Yan, and G. Wang, "Distinguishing the impacts and gradient effects of climate change and human activities on vegetation cover in the weihe river basin, china", *Journal of Geophysical Research: Biogeosciences* **129**, e2024JG008297 (2024).
- ⁵B. Li, W. Lv, H. Ding, Y. Song, and W. Zhao, "New-type urbanization policy, urban expansion, and employment quality in chinese cities: evidence from a quasi-natural experiment in China", *GIScience & Remote Sensing* **63**, 2716413 (2026).
- ⁶L. Xu, H. Shi, W. Lyu, Y. Zhang, and H. Zhao, "Beyond correlation: unveiling the causal link between street-scapes and recreational walking/cycling", *Journal of Transport Geography* **136**, 104789 (2026).
- ⁷C. Chen, Y. Song, W. Lv, A. Shemery, K. Hampson, W. Yi, Y. Zhong, and P. Wu, "Predicting pavement cracking performance using laser scanning and geocomplexity-enhanced machine learning", *Computer-Aided Civil and Infrastructure Engineering*, 10.1111/mice.13489 (2025).

Note: Publications under "Lyu, W." and "Lv, W." both refer to Wenbo Lyu.